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SUB CODE: DSA0502

SUB NAME: QUERY PROCESSING FOR DATA SCIENCE

CO5 ASSESSMENT TOOL 1

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2. Sales Performance and Trend Dashboard

Create a dashboard for an e-commerce company using sales data containing date, product, category, quantity, revenue, and region. Analyze the evolution of sales over time, product-wise distributions, and category frequencies. Include appropriate Matplotlib/Pandas plots and use a lag plot and autocorrelation plot to identify time-dependent sales patterns.

AIM: To create a sales performance and trend dashboard for an e-commerce company by analyzing sales data containing date, product, category, quantity, revenue, and region. The analysis focuses on sales trends over time, product-wise sales distribution, category frequencies, regional performance, and time-dependent sales patterns using lag and autocorrelation analysis**.**

TOOLS USED: Python ,Pandas,NumPy,Matplotlib,Seaborn,VS Code.

DATASET USED: sales_data.csv .

DATA EXPLORATION:

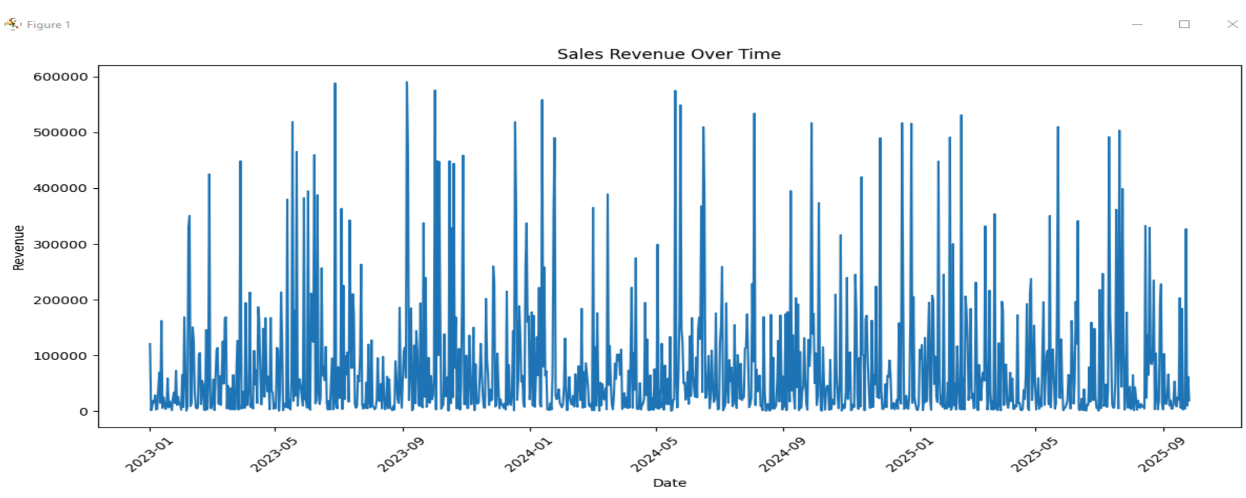
The date column was converted into an appropriate date-time format for time-based analysis.

```
Columns:
['Date', 'Product', 'Quantity', 'Region', 'Category', 'Revenue']

First 5 rows:
        Date     Product  Quantity Region     Category    Revenue
0 2023-01-01      Tablet         6   West  Electronics  121013.69
1 2023-01-02    Keyboard         1  North  Accessories    1527.10
2 2023-01-03       Mouse         9   East  Accessories    7212.04
3 2023-01-04      Tablet         1  South  Electronics   19190.19
4 2023-01-05  Headphones         5   West  Accessories   15195.07

Dataset saved as:
sales_data.csv
```

SALES TREND ANALYSIS:

Sales revenue was grouped according to date to study the evolution of sales over time.

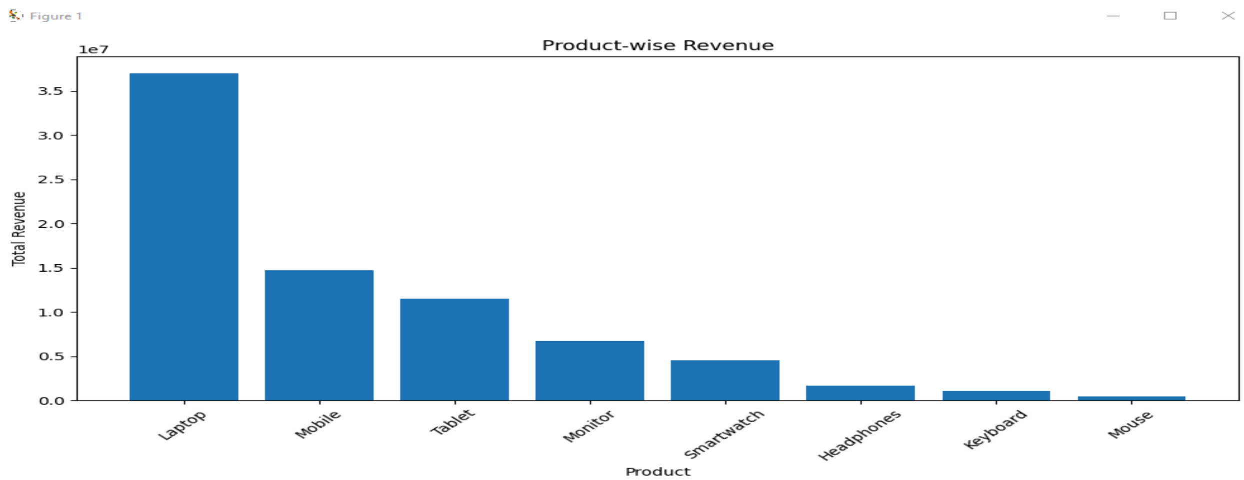
A line chart was created to visualize daily revenue and identify increases, decreases, and fluctuations in sales.



PRODUCT-WISE SALES ANALYSIS:

Sales were grouped according to individual products.

The following parameters were analyzed: Total revenue ,Total quantity sold



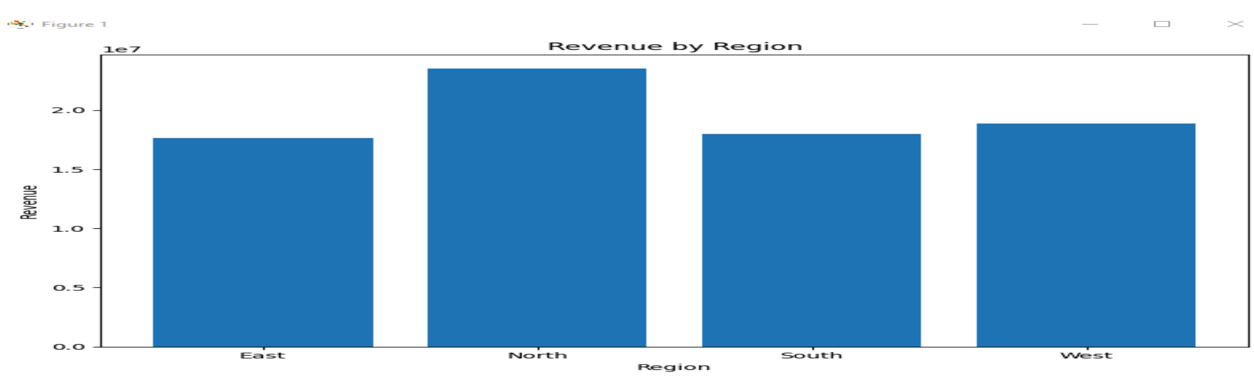
REGIONAL SALES ANALYSIS:

Sales were grouped according to region to compare regional performance.

The total revenue and quantity sold were analyzed for:

- North
- South
- East
- West

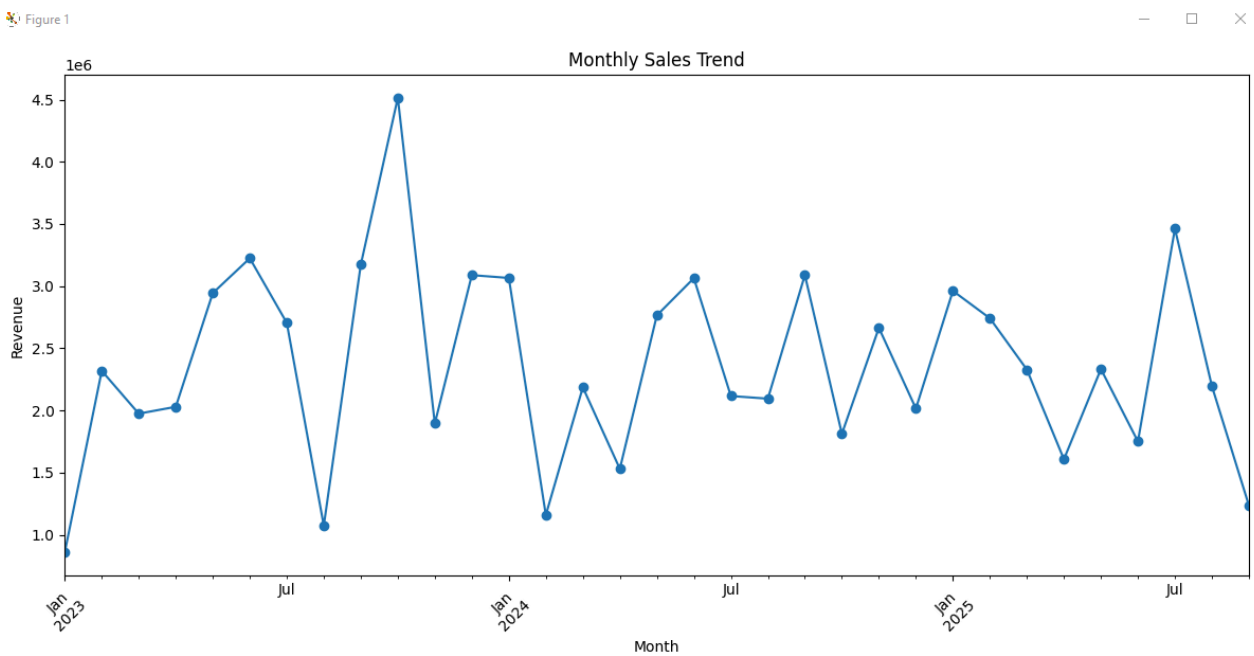
A bar chart was created to visualize regional revenue.



MONTHLY SALES TREND:

The sales data was grouped by month to understand monthly changes in revenue.

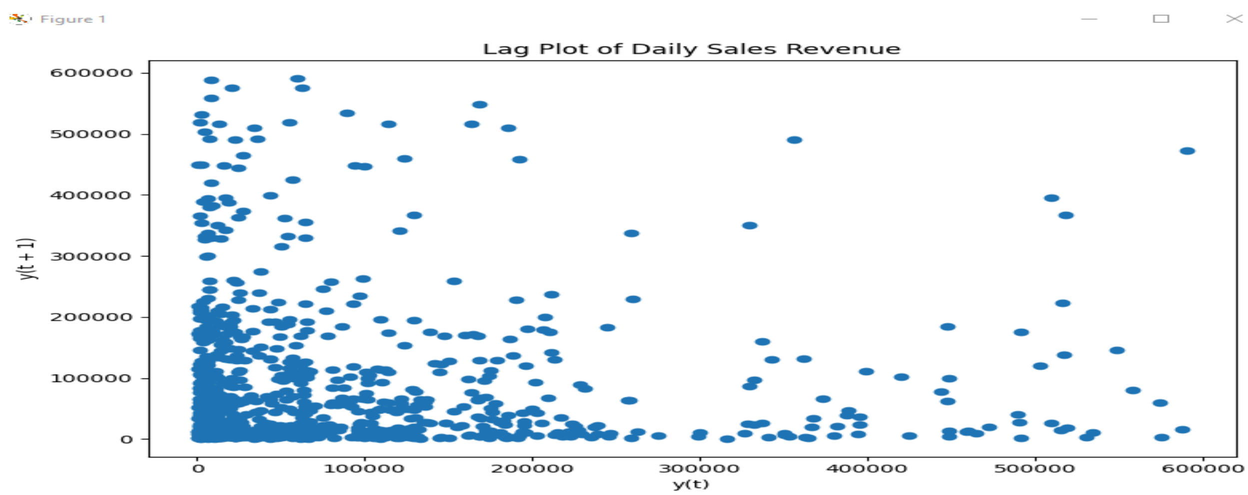
A line chart was created to identify the month with the highest and lowest sales and to observe overall sales trends.



LAG PLOT ANALYSIS:

A **lag plot** was created using daily sales revenue.

The lag plot compares the current day's sales with the previous day's sales. It was used to determine whether there is a relationship between consecutive sales observations. A visible pattern in the lag plot may indicate that previous sales values influence or are related to current sales values.

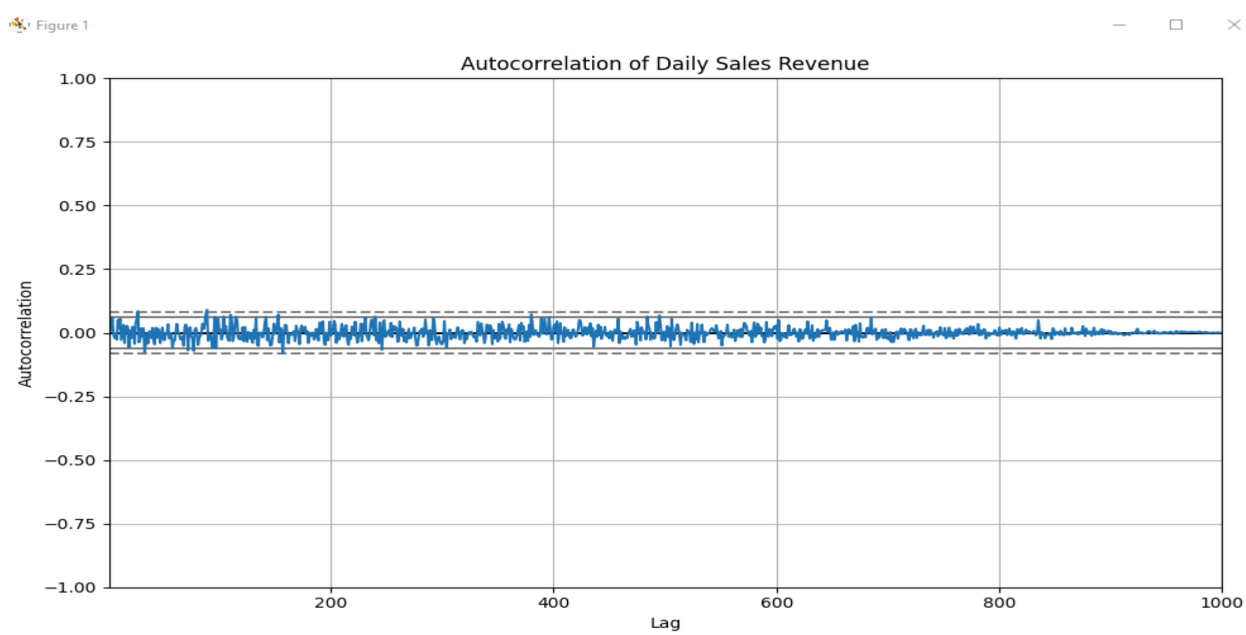


AUTOCORRELATION ANALYSIS:

An **autocorrelation plot** was created to study time-dependent patterns in daily sales.

Autocorrelation measures the relationship between a time series and its previous observations at different time lags.

This analysis helps determine whether sales show dependence or repeating patterns over time.



CONCLUSION:

The e-commerce sales dataset was successfully analyzed using Python, Pandas, NumPy, Matplotlib, and Seaborn. Sales trends were studied over time, while product-wise, category-wise, and regional sales distributions were analyzed using appropriate grouping techniques.

Various visualizations were created to understand sales performance. In addition, lag plot and autocorrelation analysis were performed to identify time-dependent relationships in sales data.

Overall, the analysis demonstrates how data analysis and visualization can help an e-commerce company understand sales performance, identify important products and categories, compare regional performance, recognize sales trends, and support data-driven business decisions.

4.Hospital Patient Analysis Dashboard

Build a dashboard using patient data containing age, blood pressure, cholesterol, BMI, treatment cost, and admission date. Present distributions, counts, and relationships between health-related variables using Matplotlib. Include time-based plots to analyze admissions and use lag/autocorrelation plots to identify recurring admission patterns.

AIM: To build a hospital patient analysis dashboard using patient data containing age, blood pressure, cholesterol, BMI, treatment cost, and admission date. The analysis focuses on studying patient distributions, admission counts, relationships between health-related variables, treatment costs, and time-based admission patterns using suitable visualizations, lag plots, and autocorrelation analysis.

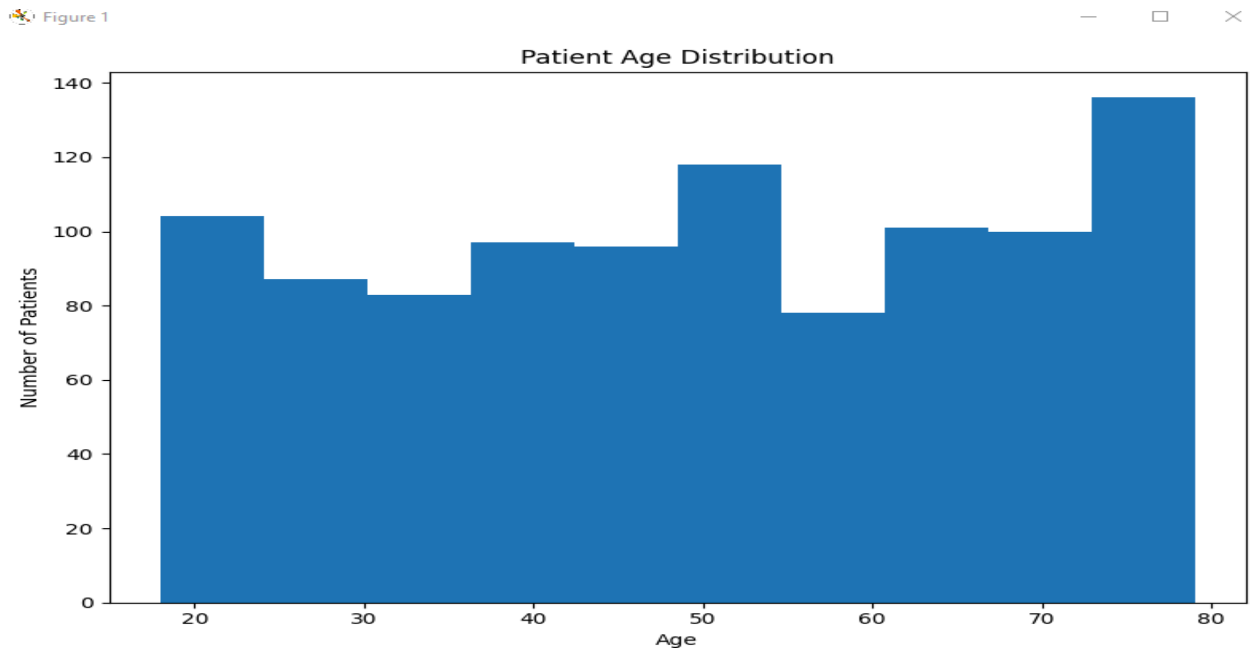
TOOLS USED: Python ,Pandas,NumPy,Matplotlib,Seaborn,VS Code.

DATASETS USED: hospital_patients.csv .

PATIENT AGE DISTRIBUTION:

Patient ages were analyzed to understand the distribution of patients across different age groups.

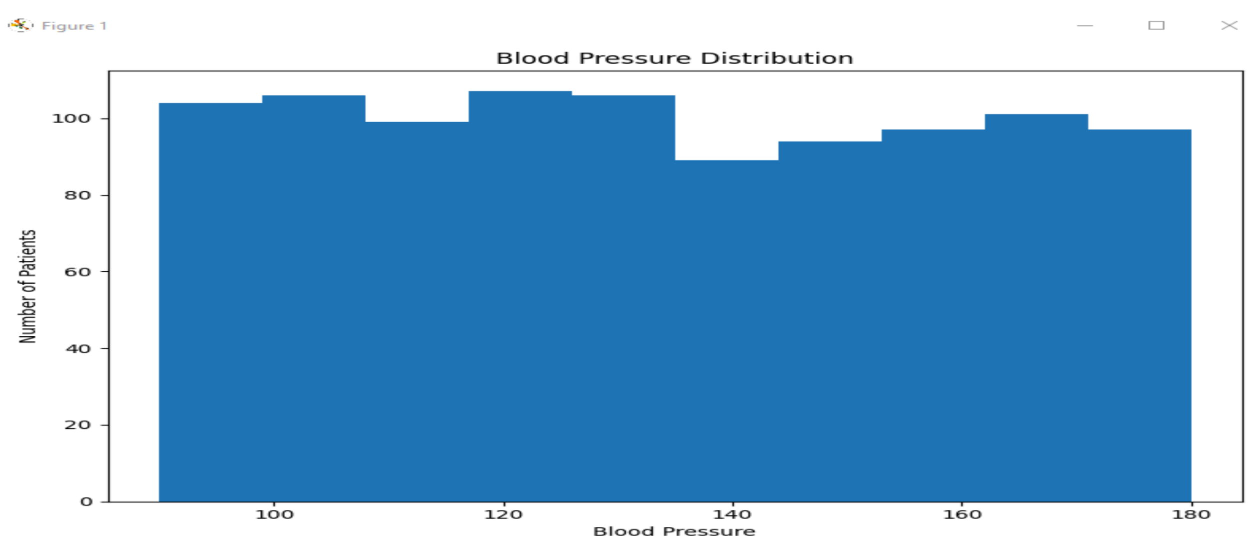
A histogram was created to visualize the frequency of patients in different age ranges.



BLOOD PRESSURE ANALYSIS:

Blood pressure values were analyzed to understand the distribution of blood pressure among patients.

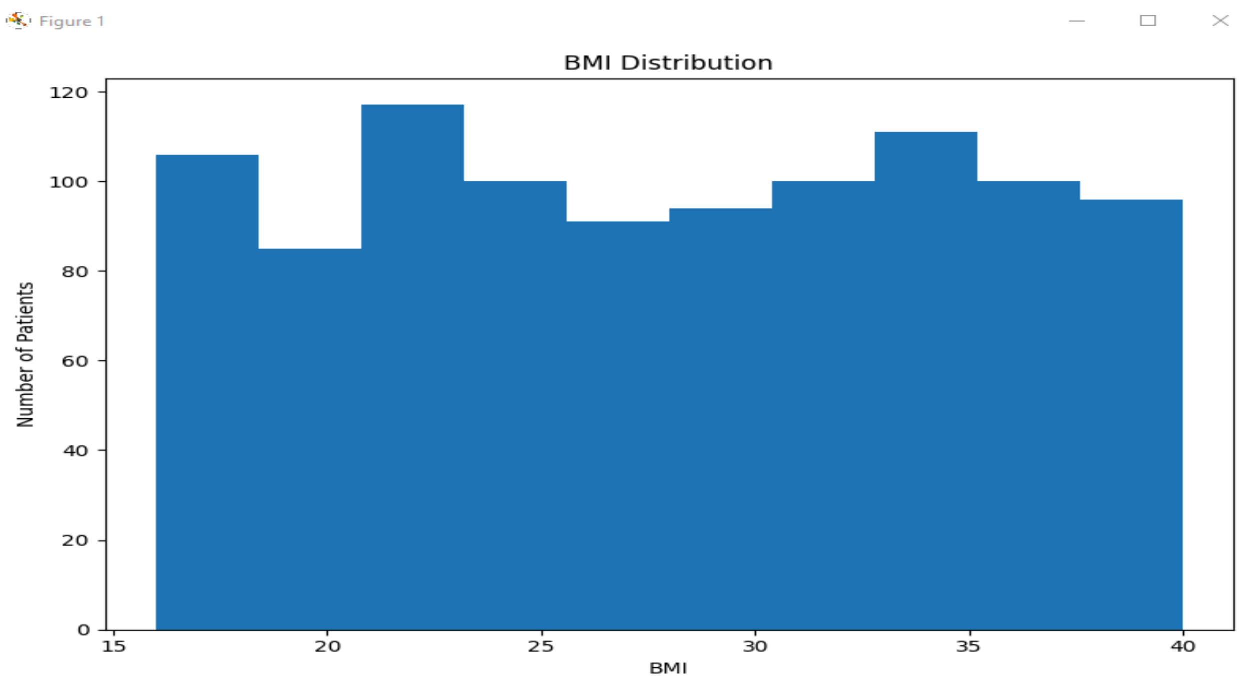
A histogram was used to visualize the distribution and identify unusually high or low values.



BMI ANALYSIS:

BMI values were analyzed to understand the distribution of patients based on their body mass index.

A histogram or box plot was used to visualize BMI distribution and identify potential unusual observations.

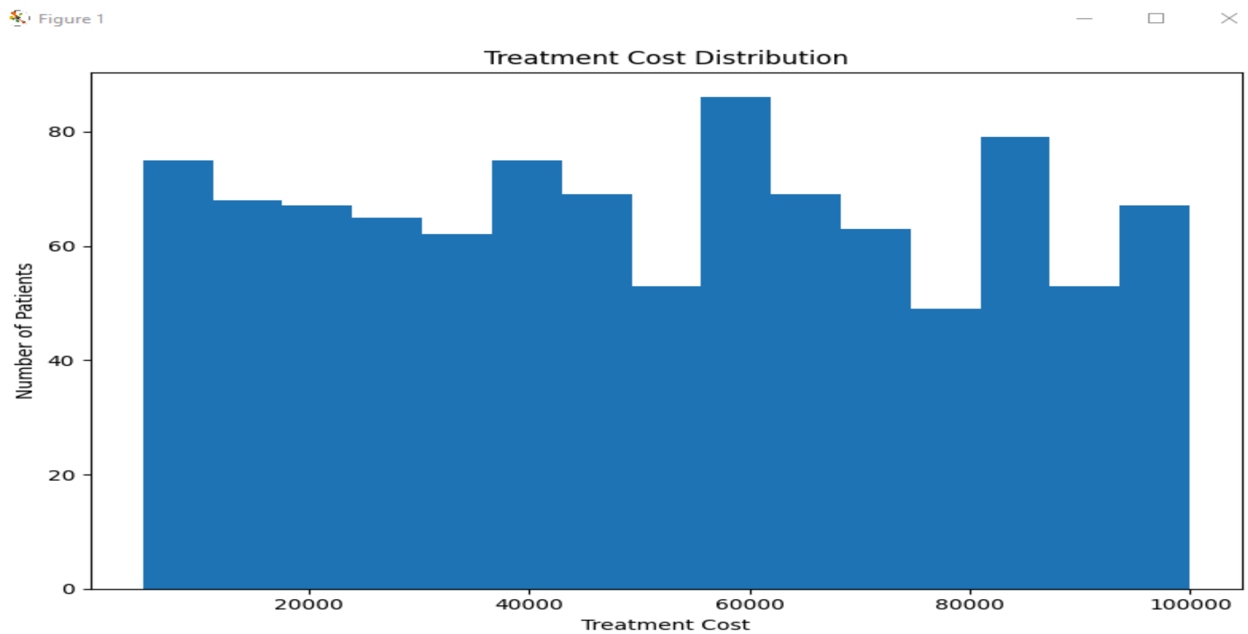


TREATMENT COST ANALYSIS

Treatment costs were analyzed to understand the financial distribution of hospital treatments.

The average, minimum, maximum, and overall distribution of treatment costs were calculated.

A histogram or box plot was created to visualize treatment costs.



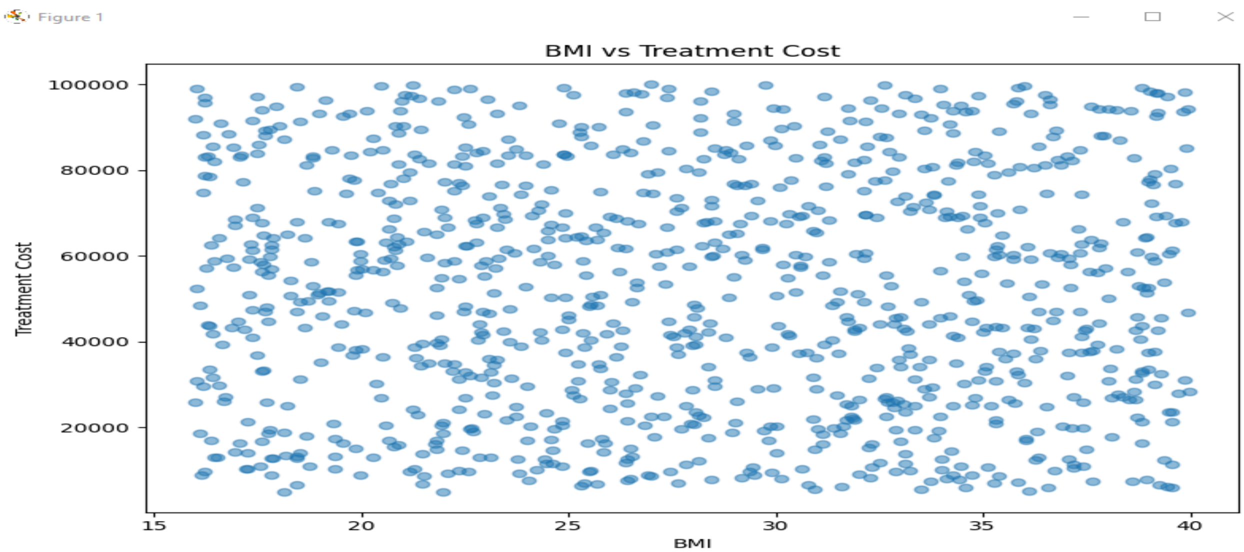
RELATIONSHIPS BETWEEN HEALTH VARIABLES

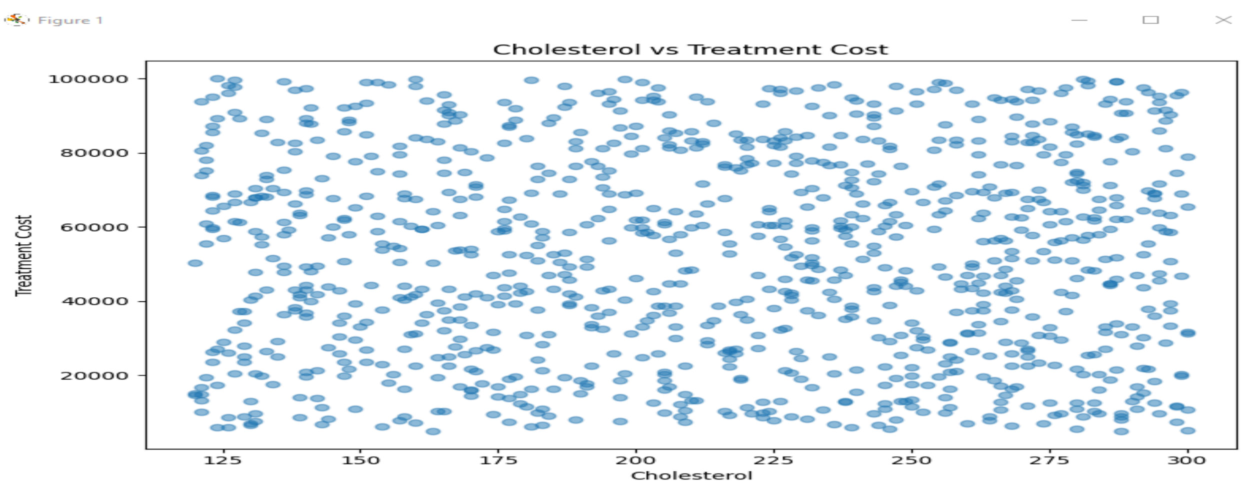
Relationships between health-related variables were analyzed using scatter plots.

The following relationships were studied:

- Age vs Blood Pressure
- Age vs Cholesterol
- BMI vs Treatment Cost
- Blood Pressure vs Treatment Cost
- Cholesterol vs Treatment Cost

These visualizations help identify possible relationships between patient health characteristics and treatment costs.



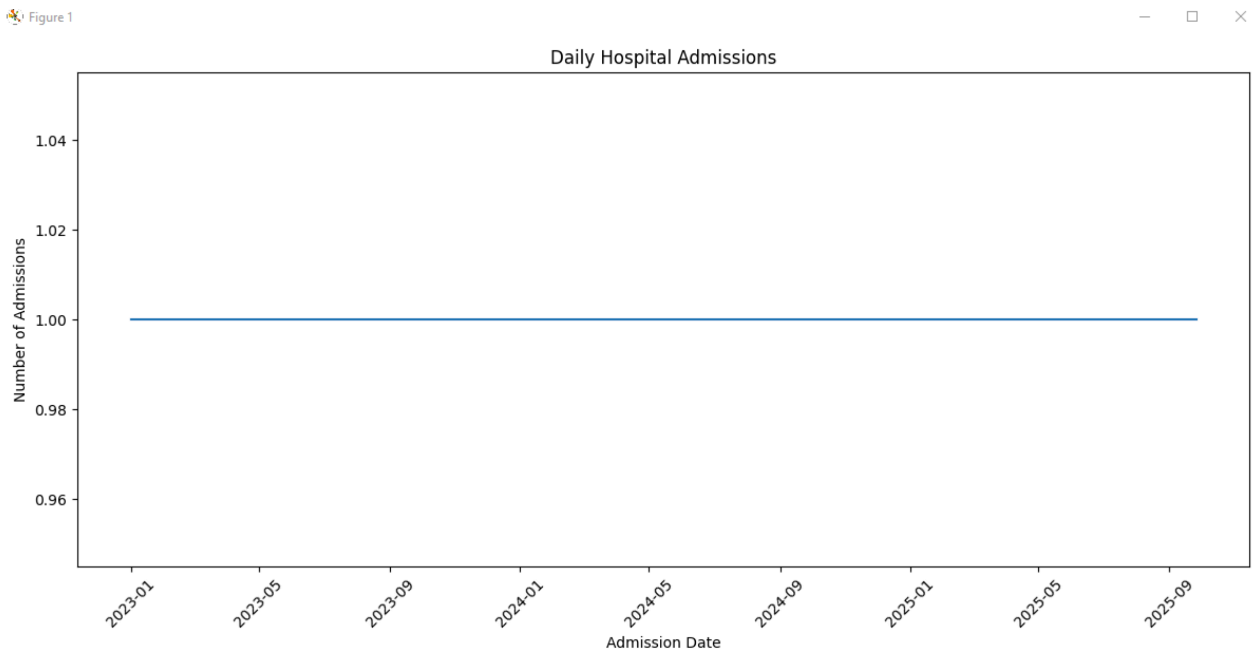


ADMISSION COUNT ANALYSIS:

Patient admissions were grouped by date to calculate the number of admissions per day.

A line chart was created to visualize the number of hospital admissions over time.

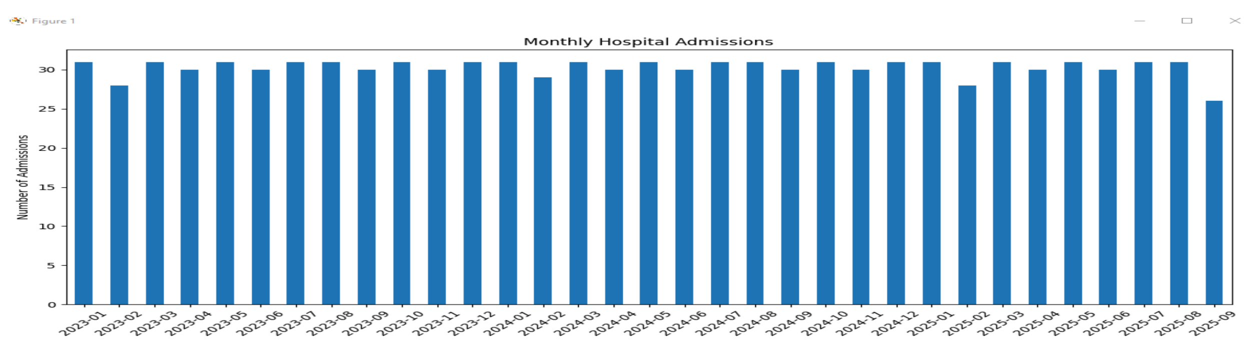
This helps identify periods with higher or lower admission activity.



MONTHLY ADMISSION ANALYSIS:

The admission dates were grouped by month to analyze monthly admission patterns.

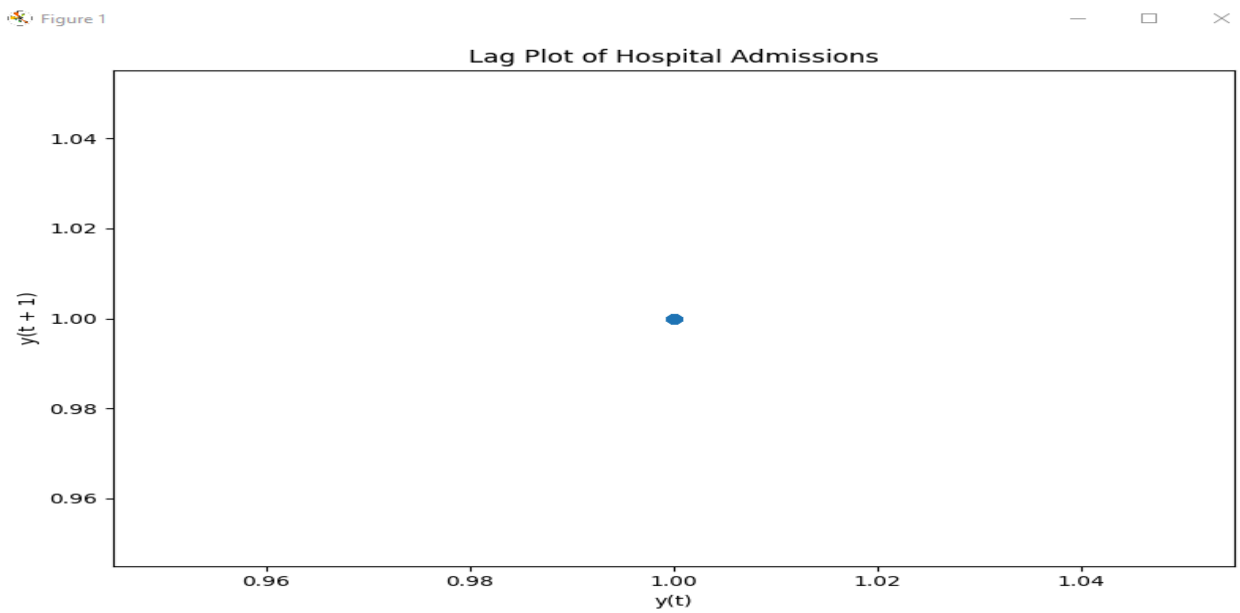
A monthly line chart was created to identify months with higher or lower patient admissions.



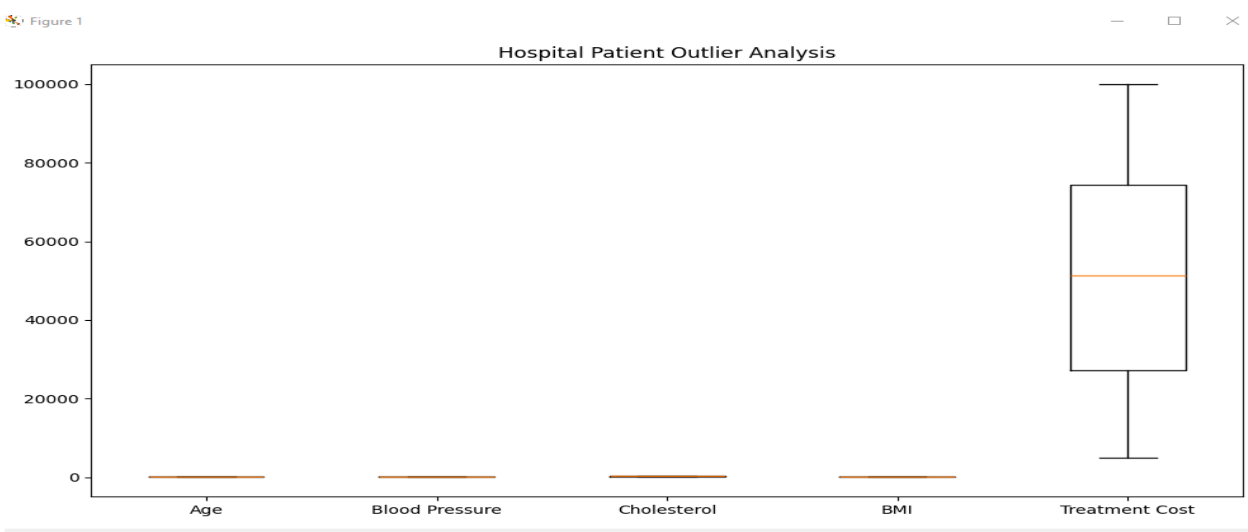
LAG PLOT ANALYSIS:

A **lag plot** was created using daily admission counts.

The lag plot compares the number of admissions on one day with the number of admissions on the previous day. A visible pattern may indicate that admissions on consecutive days are related.



OUTLIERS:

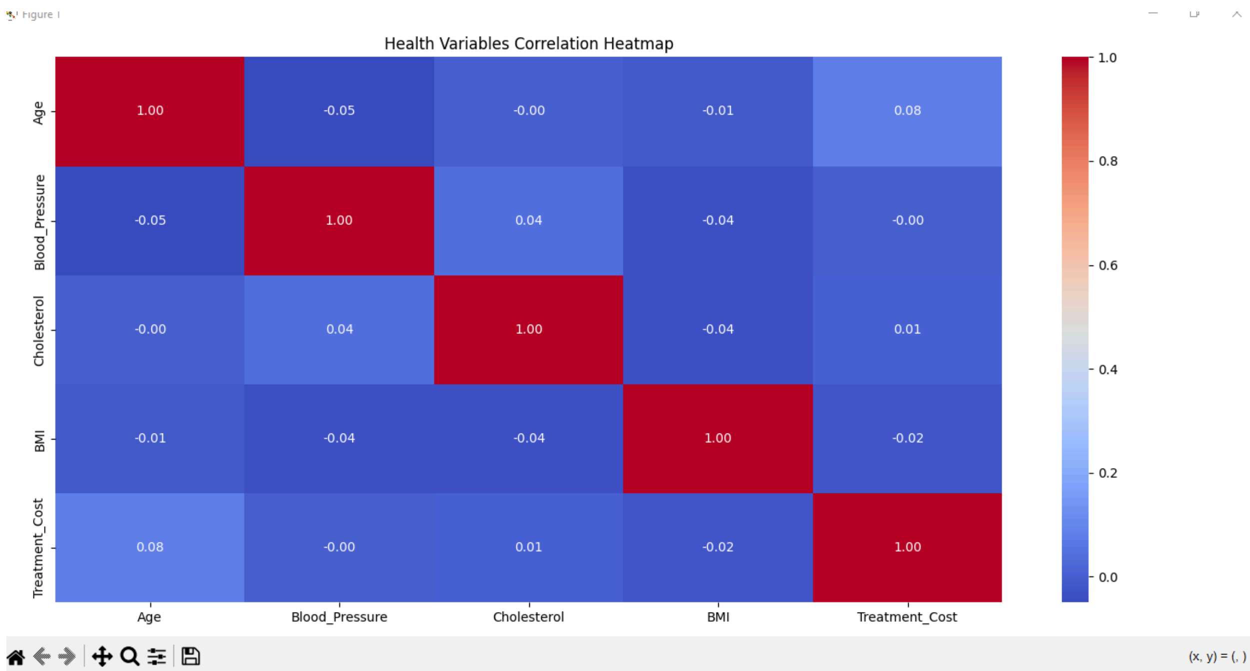


AUTOCORRELATION ANALYSIS

An **autocorrelation plot** was created to analyze whether hospital admissions show time-dependent or recurring patterns.

Autocorrelation measures the relationship between admission counts and their previous observations at different time lags.

This analysis helps identify possible recurring admission patterns.



KEY FINDINGS:

1. STRONGEST HEALTH VARIABLE RELATIONSHIP

Age and Treatment_Cost have the strongest correlation of 0.08.

2. AVERAGE PATIENT AGE

The average patient age is 49.86 years.

3. AVERAGE TREATMENT COST

The average treatment cost is ₹51,533.43.

4. HIGHEST ADMISSION MONTH

2023-01 recorded the highest number of hospital admissions with 31 admissions.

5. OUTLIER ANALYSIS

A total of 0 potential outlier observations were identified.

CONCLUSION:

The hospital patient dataset was successfully analyzed using Python, Pandas, NumPy, Matplotlib, and Seaborn. Patient characteristics such as age, blood pressure, cholesterol, BMI, and treatment cost were explored using distributions and statistical analysis.

Relationships between health-related variables were visualized using appropriate plots. Time-based analysis was performed to study daily and monthly admission trends. Lag and autocorrelation plots were also used to identify possible recurring and time-dependent admission patterns.

Overall, the analysis demonstrates how data analysis and visualization can help hospitals understand patient characteristics, treatment costs, admission trends, and recurring patterns, supporting better healthcare resource planning and decision-making.